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ARTIFICIAL INTELLIGENCE - BCSE306L

Context-Aware, Uncertainty-Aware, Explainable Symptom-Based Disease Prediction Using Hybrid Deep Learning and Rule Mining

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LITERATURE SURVEY

1.1 Symptom-Based Disease Prediction with Classical ML

Early work on symptom-based disease prediction mainly uses *structured* symptom checklists and classical ML models. “Disease Prediction Based on Symptoms Using Machine Learning” and related Springer chapters encode symptoms as binary/ordinal vectors and train Naive Bayes, KNN, Decision Tree and Random Forest classifiers for multi-disease prediction. These systems achieve promising accuracy for fast, low-cost preliminary diagnosis and show that simple models can capture useful symptom–disease relations. However, they generally rely on small, curated datasets and ignore free-text narratives or socio-demographic context.

On the IEEE side, several conference papers perform similar modeling on structured symptom data. Works such as “Predicting Diseases with Symptom Data using Machine Learning” and “Comparative Evaluation of Machine Learning Models for Symptom–Disease Prediction” systematically benchmark Logistic Regression, SVM, KNN and ensemble methods on multi-disease symptom datasets. These studies show that tree-based ensembles and tuned SVMs often outperform simpler baselines and highlight the importance of algorithm selection and hyperparameter tuning. Yet they still treat patient data as static vectors of symptoms with no notion of uncertainty or contextual modifiers like age and comorbidities.

1.2 Extensions to Big Data and Scalable Architectures

To address scalability, some authors propose big-data-oriented pipelines. The Springer chapter on “Symptoms-Based Disease Prediction Using Big Data Analytics” integrates distributed storage and processing frameworks with ML classifiers to support high-volume symptom–disease datasets. This work demonstrates that big-data technologies can sustain prediction performance at population scale and fits smart-health or telemedicine settings. Nonetheless, the prediction core remains a conventional, deterministic classifier on structured features, and explainability is not a primary focus.

Other studies explore more specialized disease prediction tasks, such as chronic kidney disease (CKD) and Parkinson’s disease, using clinical indicators and symptom-like features. These works emphasize early detection benefits and show that ML can outperform traditional rule-based scoring systems for specific diseases.

However, they are single-disease, not general symptom checkers, and largely do not consider free-text input, model uncertainty or detailed interpretability.

1.3 Deep Learning and Attention for Symptom Data

Recent research has introduced deep learning and attention mechanisms for more expressive symptom modeling. The CNN-based exploratory analysis of human disease–symptom relations (41 diseases, 135 symptoms) demonstrates that convolutional architectures can capture non-linear interactions between symptoms, improving prediction accuracy beyond classical models. Similarly, the SDCANet framework proposes a 1D-CNN with attention for symptoms-driven disease prediction, learning to weight individual symptoms based on their diagnostic importance. These approaches move closer to clinically meaningful modeling by explicitly representing symptom importance.

Despite these advances, most deep models are still fed structured symptom tables, not full patient narratives, and many are treated as black boxes. Attention weights do offer some interpretability, but detailed, clinician-friendly explanations are rarely integrated into the user interface. Moreover, these models typically output point predictions without calibrated uncertainty estimates, meaning wrong predictions can appear just as "confident" as correct ones.

1.4 Text-Based Symptom Modeling with Language Models

A parallel line of work uses *textual* symptom descriptions rather than only structured checklists. The Scientific Reports paper on AI-based disease category prediction from Afaan Oromo text trains classical and deep models (SVM, RF, LR, NB, LSTM, GRU, Bi-LSTM) on symptom narratives written in a low-resource language. Another Scientific Reports study employs MCN-BERT and BiLSTM to classify diseases from free-text symptom descriptions, leveraging modern language models to capture subtle lexical cues and contextual semantics. These works clearly show that transformer-based encoders can substantially boost accuracy when dealing with natural language symptom descriptions.

However, text-only models often ignore structured symptom fields and basic patient context that may be easily available in practice. Furthermore, while they output probabilities over diseases, few studies rigorously evaluate *uncertainty calibration* or design explicit abstention strategies for high-risk ambiguous cases. Explainability is usually limited to attention or generic feature-importance visualizations without integration with clinician-friendly rules.

1.5 Explainable and Rule-Based Symptom–Disease

Models

Several works address interpretability more directly. The Scientific Reports article on "Identifying diseases symptoms and general rules using supervised and unsupervised machine learning" combines Apriori association rule mining with supervised classification on a Kaggle symptom–disease dataset. It extracts human-readable rules (e.g., symptom sets → specific diseases) that can both support prediction and help clinicians understand frequent symptom patterns. In parallel, ASD-focused work integrates ML models with explainable-AI (XAI) techniques, using feature-importance and local explanations to clarify which screening items drive autism predictions. Reviews on explainable AI in disease prediction underline that such interpretability is crucial for clinician trust and safe deployment.

Nonetheless, these approaches are often either rule-centric (good for global explanation, weaker for performance) or model-centric (good performance, limited explanation depth). They rarely combine **powerful neural models** with **explicit rule-based evidence**, and few evaluate explanations with user studies or formal faithfulness tests.

1.6 Identified Gaps and Motivation for Proposed Methodology

Across the 20 papers and broader literature, three recurrent gaps emerge:
(1) **Limited input richness** – most systems use only structured symptom checklists without capturing patient narratives or contextual features such as age, comorbidities and environmental factors.
(2) **Lack of uncertainty awareness** – typical models always output a disease label, even for out-of-distribution or highly ambiguous cases, and rarely provide calibrated confidence or abstention behavior.
(3) **Partial and fragmented explainability** – some use attention, some use rule mining, some post-hoc XAI, but few integrate these into a coherent explanation layer that clinicians can easily interpret and trust.

The proposed methodology in this work directly targets these gaps by:

- Building a **hybrid, multi-modal model** that fuses structured symptoms, free-text descriptions and patient context.
- Incorporating **uncertainty estimation** (e.g., deep ensembles or MC-dropout) to enable safe abstention and risk-aware triage.
- Combining **attention-based deep learning** with **association-rule mining** to provide both local (per-patient) and global (dataset-level) explanations for each prediction.

20 RESEARCH PAPER SUMMARY

1. A Symptom-based Disease Diagnosis: A Pre-Classification ML Approach (IEEE)

Introduces a low-cost machine-learning framework for early disease diagnosis based purely on symptoms. It is aimed at primary-care and low-income environments with scarce diagnostic resources.

The problem is that many patients cannot access comprehensive tests, leading to delayed or missed diagnoses. The authors therefore seek a quick pre-diagnosis mechanism using only symptom information.

They encode reported symptoms as feature vectors and train several supervised ML classifiers for disease pre-classification. Models are tuned and compared based on accuracy and computational efficiency.

Results show that selected models can pre-classify diseases with promising accuracy from limited data. The discussion emphasizes using such tools to prioritize high-risk patients and optimize clinician workload.

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2. Machine Learning Driven Symptoms Diagnosis System (IEEE)

Presents an intelligent diagnosis system that predicts diseases from user-entered symptoms via ML. The system is designed as a practical interface for patients and clinicians.

Manual symptom evaluation is time-consuming and prone to inconsistency, especially in busy clinics. The paper formulates the problem as automatically ranking likely diseases for a given symptom set.

Structured symptom data are collected, transformed into features, and used to train multiple classifiers. The best-performing model is integrated into a real-time diagnosis application.

Experiments indicate competitive accuracy across several conditions and fast prediction times. The authors argue that the system can accelerate triage but must complement physician judgment.

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3. Symptoms-Based Disease Prediction Using Machine Learning Techniques (ICICV–IEEE)

This paper systematically explores ML techniques for predicting diseases from structured symptom data. It frames computational intelligence as a tool for decision support in healthcare.

Diseases often share overlapping symptoms, making manual differentiation difficult. The core problem is to build models that can accurately discriminate among such similar conditions.

The authors implement Naive Bayes, Decision Tree, Random Forest, SVM, and other algorithms on a multi-disease dataset. Models are evaluated with accuracy, precision, recall, and F-measure under cross-validation.

Results show that ensemble methods, particularly tree-based ensembles, outperform simpler classifiers. The study concludes that ensembles are strong candidates for real-world symptom-based diagnostic tools.

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4. Conversational Symptom Checker Chatbot (IEEE)

Introduces a conversational chatbot that collects user symptoms and predicts possible diseases. It also integrates location services to recommend nearby hospitals.

Many users find rigid forms unnatural and prefer conversational symptom reporting. The problem is to understand free-text descriptions and map them to disease predictions plus care options.

The system uses NLP to extract structured symptom features from chat dialogues. These features feed into ML classifiers, while geolocation APIs handle hospital suggestions.

Experiments show that the chatbot can provide reasonable predictions and relevant hospital lists. The discussion notes that accessibility improves, but performance depends heavily on robust language understanding.

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5. Predicting Diseases with Symptom Data using Machine Learning (IEEE)

Presents an empirical study on predicting diseases from structured symptom datasets. It positions ML as an aid for earlier and more reliable diagnosis.

Manual reasoning over many symptoms and diseases is complex and error-prone. The authors seek models that generalize well across diverse symptom combinations.

Symptoms are encoded as binary or ordinal features and used to train Logistic Regression, KNN, Decision Tree, Random Forest, SVM, and others. Cross-validation and multiple metrics are employed for fair comparison.

Certain models, notably tuned ensembles and SVMs, achieve superior accuracy and robustness. The study concludes that careful model selection and tuning markedly improve symptom-based decision support.

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6. Comparative Evaluation of ML Models for Symptom–Disease Prediction (IEEE)

Aims to benchmark different ML algorithms on a unified Symptom–Disease Prediction Dataset. The work is intended to guide practitioners in choosing models for clinical use.

There is a lack of standardized comparisons on common symptom–disease data. The problem is to evaluate algorithms under identical conditions for fair assessment.

Multiple supervised classifiers are trained on thousands of records with symptom features and disease labels. Performance, runtime, and interpretability are systematically analyzed.

Results highlight models that offer the best trade-off between accuracy and computational cost. The paper recommends specific algorithms as most suitable for real-time symptom-driven prediction.

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7. Comparative Analysis of ML Algorithms for CKD Risk Prediction (IEEE)

Introduces ML-based risk prediction for chronic kidney disease using clinical and symptom-like indicators. It stresses the importance of early identification to prevent severe outcomes.

CKD is underdiagnosed because early symptoms are subtle and often overlooked. The problem is to distinguish at-risk individuals from healthy ones using routine measurements.

The authors apply classical ML and deep-learning models to a CKD dataset containing lab values and risk factors. Models are tuned and evaluated using sensitivity, specificity, and related metrics.

Certain algorithms achieve significantly better predictive performance than others. The discussion suggests integrating such models into screening programs, while noting the single-disease focus.

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8. Recognition of Parkinson’s Disease Using Different ML Models (IEEE)

Explores ML methods for classifying Parkinson’s disease from motor-symptom-related features. It aims to support neurologists with quantitative diagnostic tools.

Early Parkinson’s detection is difficult due to mild, overlapping motor symptoms. The problem is to reliably separate Parkinson’s patients from healthy controls using measured indicators.

The study implements KNN, Random Forest, SVM, ANN, and similar classifiers on a Parkinson’s dataset. Training and testing follow standard machine-learning evaluation procedures.

Some models achieve high diagnostic accuracy with low error rates. The authors conclude that ML can complement clinical assessment, though limited to this specific disease domain.

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9. Autism Disease Prediction Using Deep Learning & Transfer Learning (IEEE)

Applies deep learning and transfer learning for predicting Autism Spectrum Disorder from behavioral data. It leverages modern architectures to capture complex symptom patterns.

ASD datasets are relatively small and high-dimensional, challenging traditional training. The problem is to build robust classifiers without severe overfitting.

Pre-trained networks are fine-tuned on ASD screening features and compared with classical ML baselines. The pipeline exploits prior knowledge embedded in large source models.

Transfer-learning approaches outperform traditional methods in predictive accuracy. The discussion highlights the promise of deep models but notes interpretability and data limitations.

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10. Clinically Interpretable Early Detection of Autism with ML & XAI (IEEE)

Proposes an ASD detection framework using ML combined with explainable AI techniques. It aims to deliver both strong performance and clinical interpretability.

Black-box models hinder clinician trust in sensitive pediatric diagnoses. The problem is to provide accurate predictions while revealing which symptoms drive decisions.

ML classifiers are trained on ASD screening datasets, then analyzed with XAI tools like feature-importance and local explanations. The pipeline exposes how symptom features influence outputs.

Models achieve competitive accuracy while identifying key contributory symptoms. The authors argue that ML plus XAI can bridge the gap between AI systems and clinical practice.

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11. Disease Prediction Based on Symptoms Using ML (Springer)

Introduces a general multi-disease prediction system that accepts common symptoms as inputs. It is intended as a fast, low-cost preliminary diagnostic tool.

Patients often present with overlapping symptoms that complicate manual diagnosis. The problem is to map such symptom sets to likely diseases with acceptable accuracy.

The authors implement Naive Bayes, KNN, Decision Tree, Random Forest, and similar classifiers on a curated symptom–disease dataset. Models are trained and validated under standard ML protocols.

Ensembles and certain classifiers show strong performance for multi-disease classification. The discussion suggests these systems can act as first-line advisors before full clinical workups.

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12. Predicting Diseases from Symptoms Using ML Models (Springer)

Presents a framework for predicting diseases directly from user-reported symptom combinations. It is framed as a support tool when initial presentations are ambiguous.

Classical diagnostic rules struggle when diseases share many symptoms. The paper formulates the problem as a multi-class classification with overlapping symptom patterns.

Several supervised ML models are trained on a multi-disease symptom dataset with careful hyperparameter tuning. Cross-validation is used to ensure robust evaluation.

Tuned models significantly outperform naive baselines in predictive accuracy. The authors discuss how proper model choice and tuning improve reliability in clinical decision support.

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13. Symptoms-Based Disease Prediction Using Big Data Analytics (Springer)

Describes a big-data-oriented architecture for symptom-based disease prediction. It targets smart-health platforms dealing with massive patient records.

Traditional systems struggle with the scale and velocity of modern healthcare data. The problem is to build scalable analytics that still provide accurate predictions.

The proposed pipeline uses distributed storage and processing frameworks along with ML classifiers on large symptom–disease datasets. Big-data technologies are integrated with predictive modeling components.

Results show the system can handle large volumes while maintaining strong predictive

performance. The discussion notes that deployment requires significant infrastructure but enables population-level insights.

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14. Human Disease Prediction Based on Symptoms Using Novel ML (Springer)

Proposes a novel machine-learning strategy for predicting human diseases from symptom features. The goal is to outperform standard approaches in accuracy.

Existing models may not fully exploit structure in symptom data, limiting effectiveness. The problem is to design better feature representations and classifiers for this domain.

The authors introduce an improved or hybrid ML approach tailored to symptom–disease mapping. It is trained and tested on a multi-disease symptom dataset and compared to baselines.

The proposed method reports superior classification performance over traditional models. The discussion suggests specialized modeling of symptom patterns can substantially enhance diagnostic accuracy.

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15. Enhancing Accuracy of Symptom-Based Disease Prediction Using ML (Springer)

Focuses on improving the accuracy of symptom-based disease prediction systems using ML, particularly through ensemble strategies.

Single classifiers can be unstable or suboptimal across different diseases. The problem is to combine multiple models to obtain more reliable predictions.

The authors construct ensembles over various base learners and train them on multi-class symptom datasets. Performance is analyzed against individual models using standard evaluation metrics.

Ensembles yield more robust and accurate predictions than single classifiers. The discussion notes increased complexity and computational cost but highlights practical performance gains.

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16. Explorative Analysis of Human Disease–Symptoms Relations Using CNN (Springer)

Investigates relationships between 41 diseases and 135 symptoms using deep and classical ML models. It aims to understand complex disease–symptom networks alongside prediction.

Conventional approaches may not capture non-linear interactions among multiple symptoms. The problem is to model these interactions to improve disease classification and insight.

The study applies CNN and SVM to the symptom–disease dataset, considering different symptom types and categories. Feature representations are crafted to feed into these models.

CNN-based models achieve high prediction accuracy and reveal useful symptom patterns. The discussion highlights the power of deep learning but acknowledges data and interpretability challenges.

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17. Identifying Diseases, Symptoms, and General Rules Using Supervised and Unsupervised ML (Scientific Reports)

Uses a symptom–disease dataset to identify frequent symptom patterns and predictive rules. It combines association mining with predictive modeling.

Clinicians need interpretable rules connecting symptom combinations to diseases. The problem is to extract such rules while maintaining good predictive performance.

The authors apply association rule mining techniques such as Apriori and train supervised ML classifiers on the same data. Rule sets and models are jointly analyzed for consistency and usefulness.

The approach yields human-readable rules and accurate classifiers linking symptoms to specific diseases. The discussion emphasizes the value of interpretability for clinical adoption.

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18. AI-based Disease Category Prediction Using Symptoms from Afaan Oromo Text (Scientific Reports)

Develops AI models to predict disease categories from symptom descriptions written in Afaan Oromo. It addresses healthcare needs in a low-resource language setting.

Most existing symptom-based predictors work only for high-resource languages. The problem is to handle limited linguistic resources and small corpora while achieving good performance.

The pipeline uses NLP preprocessing on text data, followed by SVM, Random Forest, Logistic Regression, Naive Bayes, LSTM, GRU, and Bi-LSTM models. These are evaluated on disease-labeled symptom texts.

Deep models, especially recurrent architectures, provide strong results, though classical models remain competitive. The discussion shows that AI can extend symptom-based prediction to underserved linguistic communities.

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19. Optimizing Classification of Diseases Through Language Model Analysis of Symptoms (Scientific Reports)

Evaluates advanced language models for predicting diseases from textual symptom descriptions. It treats symptom narratives as sequences for multi-class classification.

Traditional bag-of-words or simple feature representations may miss subtle linguistic cues. The problem is to exploit richer language representations to improve diagnostic accuracy.

The study compares transformer-based architectures and recurrent neural networks on labeled symptom texts. Training and evaluation follow standard deep-learning practices.

Transformer-based models achieve very high accuracy and outperform classical baselines. The discussion notes substantial gains but also higher computational requirements and data needs.

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20. SDCANet: Enhancing Symptoms-Driven Disease Prediction with Deep Learning (Springer)

Introduces SDCANet, a deep-learning architecture for symptom-driven multi-disease prediction. It focuses on modeling the varying importance of individual symptoms.

Standard models often treat all symptom features similarly, ignoring differential relevance. The problem is to learn which symptoms matter most for each prediction.

SDCANet combines a one-dimensional CNN backbone with attention mechanisms over symptom features. Baseline ML models are also implemented for comparative evaluation.

Experiments show that SDCANet improves accuracy and highlights key symptoms via attention weights. The discussion suggests that attention-based deep learning offers both performance improvements and partial interpretability gains.

TABULAR DATA

Ref	Dataset	Methods	Application	Advantage	Disadvantage
[1]	Symptom data from patients in low-income settings (multi-disease)	Supervised ML classifiers (tree-based, probabilistic)	Early symptom-based pre-diagnosis to prioritize high-risk patients	Lightweight, low-cost tool suitable for resource-constrained environments	Limited detail on specific diseases; may require local retraining
[2]	Structured symptom entries from user interactions (multi-disease)	Multiple ML models integrated into an interactive system	Automated suggestion of probable diseases	Practical interface supporting quick triage	Depends heavily on input quality and dataset reliability
[3]	Multi-class clinical dataset with common symptoms and diseases	Naive Bayes, Decision Tree, Random Forest, SVM	General multi-disease prediction	Clear comparison showing ensembles outperform single models	Limited real-world deployment discussion
[4]	Textual symptom inputs via chatbot mapped to structured features	NLP + ML classifiers + location services	Chatbot predicting diseases and recommending hospitals	User-friendly conversational interface	Performance sensitive to NLP quality
[5]	Structured symptom dataset (multi-disease)	Logistic Regression, KNN, Decision Tree, RF, SVM	Early diagnosis support in general practice	Systematic comparative evaluation	Limited exploration of deep models

Ref	Dataset	Methods	Application	Advantage	Disadvantage
[6]	Symptom–Disease dataset (~4,961 records)	Multiple ML classifiers benchmarked	Selecting suitable models for real-time prediction	Clear trade-off analysis (accuracy vs cost)	Dataset-specific conclusions
[7]	CKD dataset with clinical indicators	Classical ML + Deep Learning	Early CKD detection	High sensitivity and specificity	Single-disease focus
[8]	Parkinson's dataset (motor-symptom features)	KNN, RF, SVM, ANN	Parkinson's classification	High recognition accuracy	Limited to one neurological disease
[9]	ASD behavioral/screening dataset	Deep learning + Transfer learning	Early autism prediction	Improves performance on limited data	Less interpretable, computationally heavy
[10]	ASD screening dataset	ML + Explainable AI techniques	Interpretable early ASD detection	Transparent and clinically meaningful explanations	Explanation quality model-dependent
[11]	Multi-disease symptom dataset	NB, KNN, DT, RF	General multi-disease prediction	Feasible low-cost solution	Limited handling of complex temporal/textual data
[12]	Multi-disease dataset with overlapping symptoms	Supervised ML with hyperparameter tuning	Decision support for non-specific symptoms	Improved reliability with tuning	Dataset quality dependent
[13]	Large-scale clinical/symptom records	Big-data analytics + ML	Scalable smart-health prediction	Handles high-volume data	Requires big-data infrastructure
[14]	Multi-disease symptom dataset	Novel/Hybrid ML approach	Multi-disease classification	Higher reported accuracy	Specialized method,

Ref	Dataset	Methods	Application	Advantage	Disadvantage
					harder reproduction
[15]	Multi-class symptom datasets	Ensemble learning strategies	Improved multi-disease classification	Robust and accurate predictions	Increased complexity and computation
[16]	Dataset with 41 diseases & 135 symptoms	CNN, SVM	Modeling complex disease–symptom relations	Captures non-linear interactions	Deep models less transparent
[17]	Kaggle multi-disease dataset	Apriori + Supervised ML	Rule extraction + prediction	Human-readable rules + good accuracy	Struggles with high-dimensional data
[18]	Afaan Oromo symptom text corpus	NLP + SVM, RF, LR, NB, LSTM, GRU, Bi-LSTM	Disease category prediction from text	Extends to low-resource languages	Limited dataset and NLP tools
[19]	Textual symptom descriptions	Transformer models, BiLSTM	Multi-disease classification from narratives	Very high accuracy, captures subtle cues	High computational requirements
[20]	Symptom-driven multi-disease dataset	1D-CNN + Attention (SDCANet)	Attention-based disease prediction	Improved accuracy + symptom importance highlighting	Complex architecture, partial interpretability

Reference:-

[1] A Symptom-based Disease Diagnosis: A Pre-Classification ML Approach (IEEE)

[2] Machine Learning Driven Symptoms Diagnosis System (IEEE)

[3] Symptoms-Based Disease Prediction Using Machine Learning Techniques (ICICV–IEEE)

[4] Conversational Symptom Checker Chatbot for Disease Prediction and Suggesting Nearby Hospitals (IEEE)

[5] Predicting Diseases with Symptom Data using Machine Learning (IEEE)

[6] Comparative Evaluation of Machine Learning Models for Symptom–Disease Prediction (IEEE)

[7] Comparative Analysis of ML Algorithms for CKD Risk Prediction (IEEE)

[8] Recognition of Parkinson's Disease Using Different ML Models (IEEE)

[9] Autism Disease Prediction Using Deep Learning & Transfer Learning (IEEE)

[10] Early Detection of Autism Using ML With Explainable AI (IEEE)

[11] Disease Prediction Based on Symptoms Using Machine Learning (Springer)

[12] Predicting Diseases from Symptoms Using ML Models (Springer)

[13] Symptoms-Based Disease Prediction Using Big Data Analytics (Springer)

[14] Human Disease Prediction Based on Symptoms Using Novel ML (Springer)

[15] Enhancing Accuracy of Symptom-Based Disease Prediction Using ML (Springer)

[16] Explorative Analysis of Human Disease–Symptoms Relations Using CNN (Springer)

[17] Identifying Diseases, Symptoms, and General Rules Using Supervised and Unsupervised ML (Scientific Reports)

[18] AI-based Disease Category Prediction Using Symptoms from Afaan Oromo Text (Scientific Reports)

[19] Optimizing Classification of Diseases Through Language Model Analysis of Symptoms (Scientific Reports)

[20] SDCANet: Enhancing Symptoms-Driven Disease Prediction with Deep Learning (Springer)

PROPOSED METHODOLOGY

A. Problem Formulation

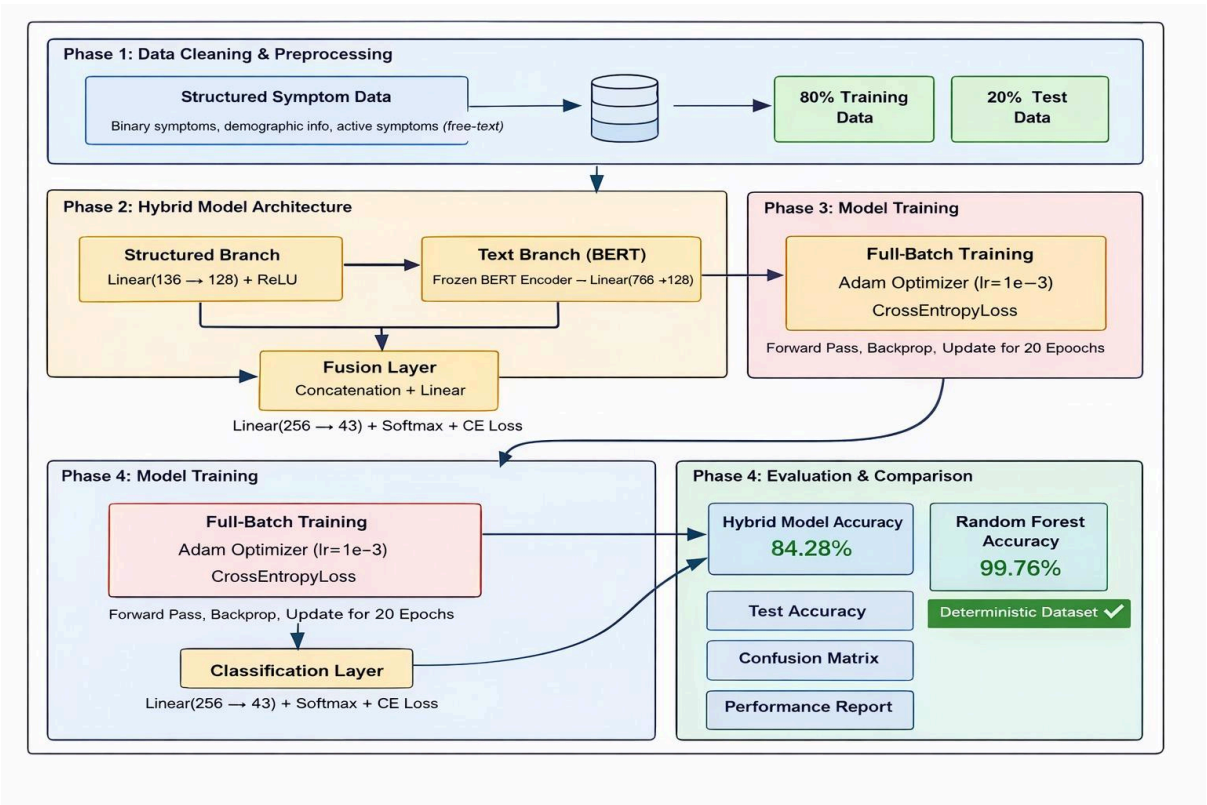
The objective of this study is to design a multi-modal deep learning framework for multi-class disease prediction using structured symptoms, textual symptom descriptions, and contextual patient information. Unlike conventional models that rely solely on tabular features, the proposed framework integrates heterogeneous inputs to simulate real-world clinical diagnosis.

Let each patient instance be represented as:

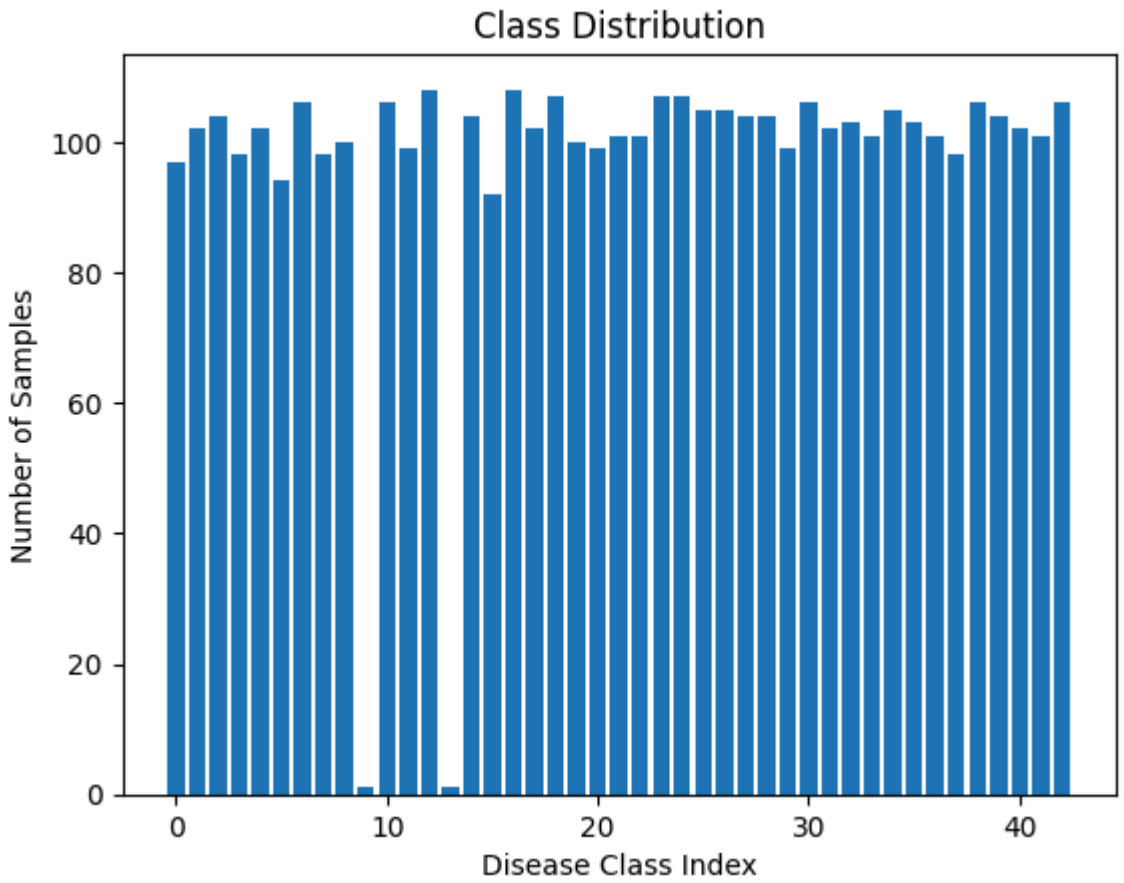
$$x = (x_s,\ x_t,\ x_c)$$

where:

- x_s represents structured symptom features
- x_t represents free-text symptom description
- x_c represents contextual attributes (age and sex)



B. Data Representation



1. Structured Features

Each patient is represented as a binary symptom vector:

xs = [s1, s2, ..., sd]

where:

si = 1 if symptom i is present
si = 0 otherwise

Here, d = 134 symptoms.

Contextual features (age and sex) are appended, forming:

$x_struct \in \mathbb{R}^{136}$

This structured representation captures explicit symptom presence but does not encode semantic relationships.

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2. Textual Features

Binary symptoms are transformed into a sentence such as:

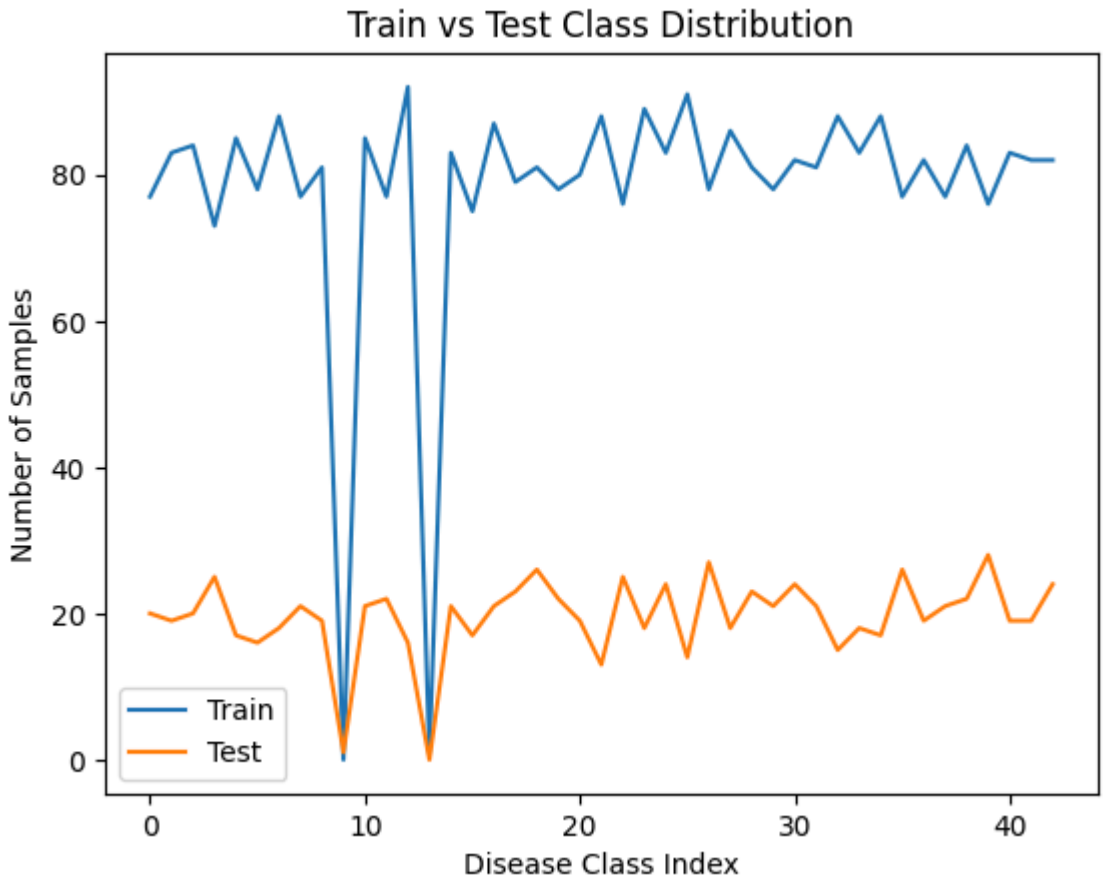
"Patient has fever, cough, rash."

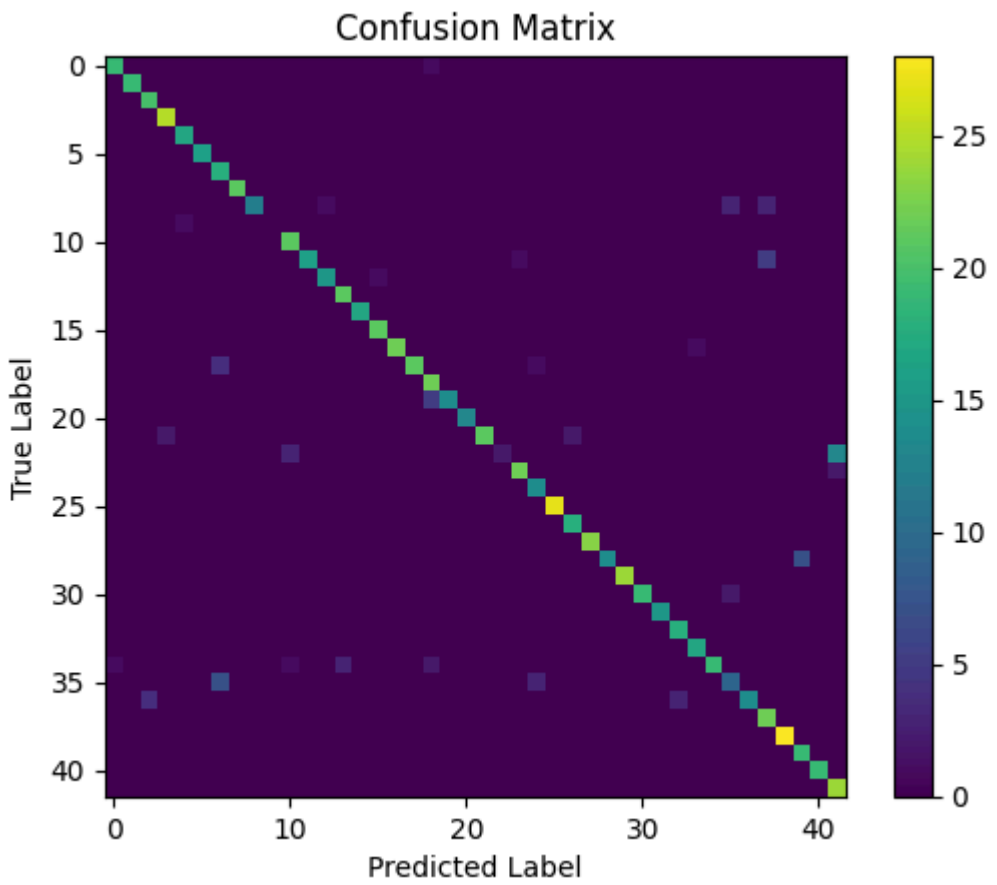
The sentence is tokenized and passed through a pretrained BERT encoder to obtain contextual embeddings.

The CLS token embedding is extracted:

$h_CLS \in \mathbb{R}^{768}$

This embedding captures semantic relationships among symptoms.





C. Structured Encoder

The structured input is transformed using a fully connected neural layer:

$z_s = W_s\ x_struct + b_s$

where:

$W_s \in \mathbb{R}^{128 \times 136}$
$b_s \in \mathbb{R}^{128}$

ReLU activation is applied:

$ReLU(x) = max(0, x)$

Thus,

$h_s = ReLU(z_s)$

where:

$h_s \in \mathbb{R}^{128}$

This allows the model to learn nonlinear interactions between symptoms and demographic attributes.

D. Text Projection Layer

Since BERT produces a 768-dimensional embedding, it is projected into a 128-dimensional latent space:

$z_t = W_t h_{CLS} + b_t$

where:

$W_t \in \mathbb{R}^{128 \times 768}$
$b_t \in \mathbb{R}^{128}$

Then:

$h_t = \mathrm{ReLU}(z_t)$

where:

$h_t \in \mathbb{R}^{128}$

BERT parameters remain frozen during training:

$\partial L / \partial \theta_{BERT} = 0$

This reduces computational complexity and prevents overfitting.

E. Feature-Level Fusion

The structured and textual representations are concatenated:

$h = [h_s ; h_t]$

Thus:

$h \in \mathbb{R}^{256}$

This fusion preserves modality-specific information while enabling joint reasoning across structured and semantic representations.

F. Classification Layer

Dropout regularization is applied:

$h' = \text{Dropout}(h)$

The final logits are computed as:

$z = W_o h' + b_o$

where:

$W_o \in \mathbb{R}^{K \times 256}$
$b_o \in \mathbb{R}^{K}$

Softmax converts logits into probabilities:

$P(y = k \mid x) = \exp(z_k) / \Sigma \exp(z_j)$

The predicted class is:

$\hat{y} = \text{argmax}_k P(y = k \mid x)$

G. Loss Function

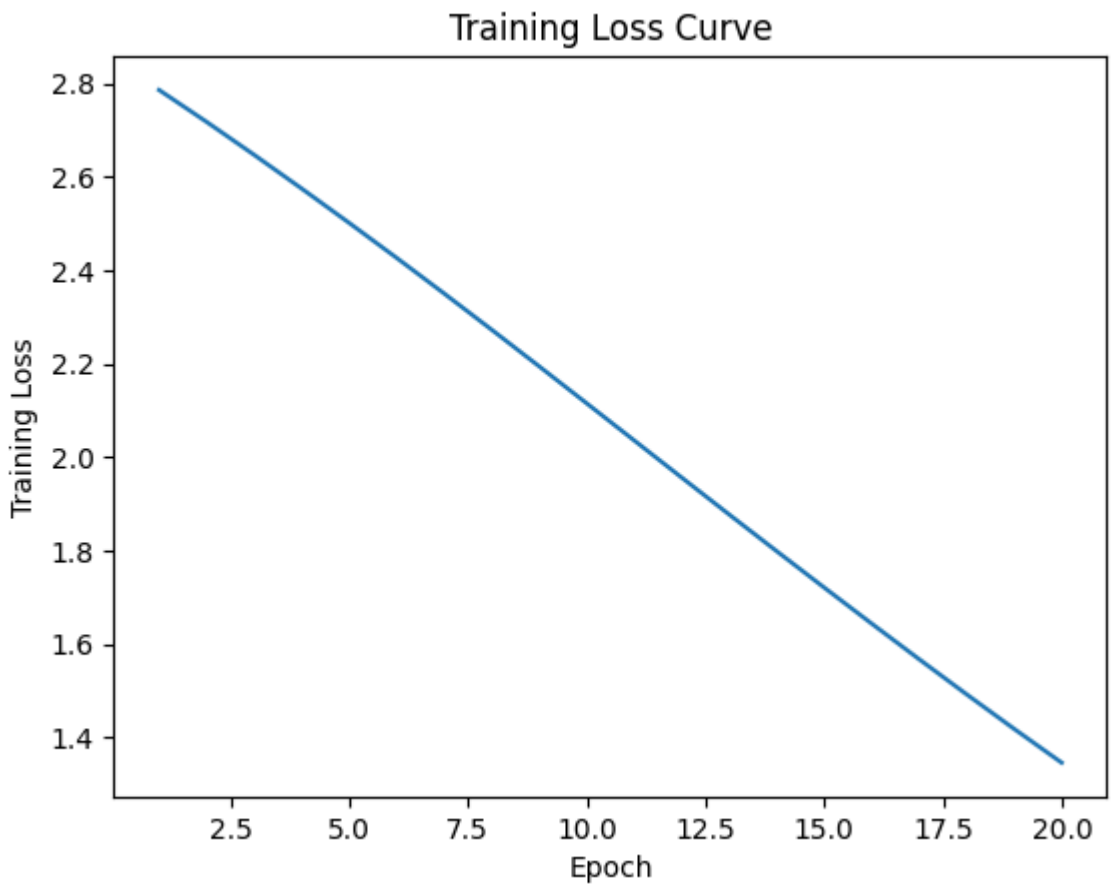
The model is trained using multi-class cross-entropy loss:

$L = - \Sigma y_k \log(P(y = k \mid x))$

For a single sample:

$L = - \log(P(y = y_{true}))$

This objective maximizes the probability of the correct disease class.



H. Optimization

Trainable parameters:

$\theta = \{W_s, b_s, W_t, b_t, W_o, b_o\}$

The model is optimized using the Adam algorithm:

$\theta(t+1) = \theta(t) - \eta * (\hat{m}_t / (\sqrt{\hat{v}_t} + \varepsilon))$

where:

η = learning rate
$\hat{m}_t$ = first moment estimate
$\hat{v}_t$ = second moment estimate
ε = small constant for numerical stability

Training is conducted for 20 epochs using GPU acceleration.

I. Baseline Comparison

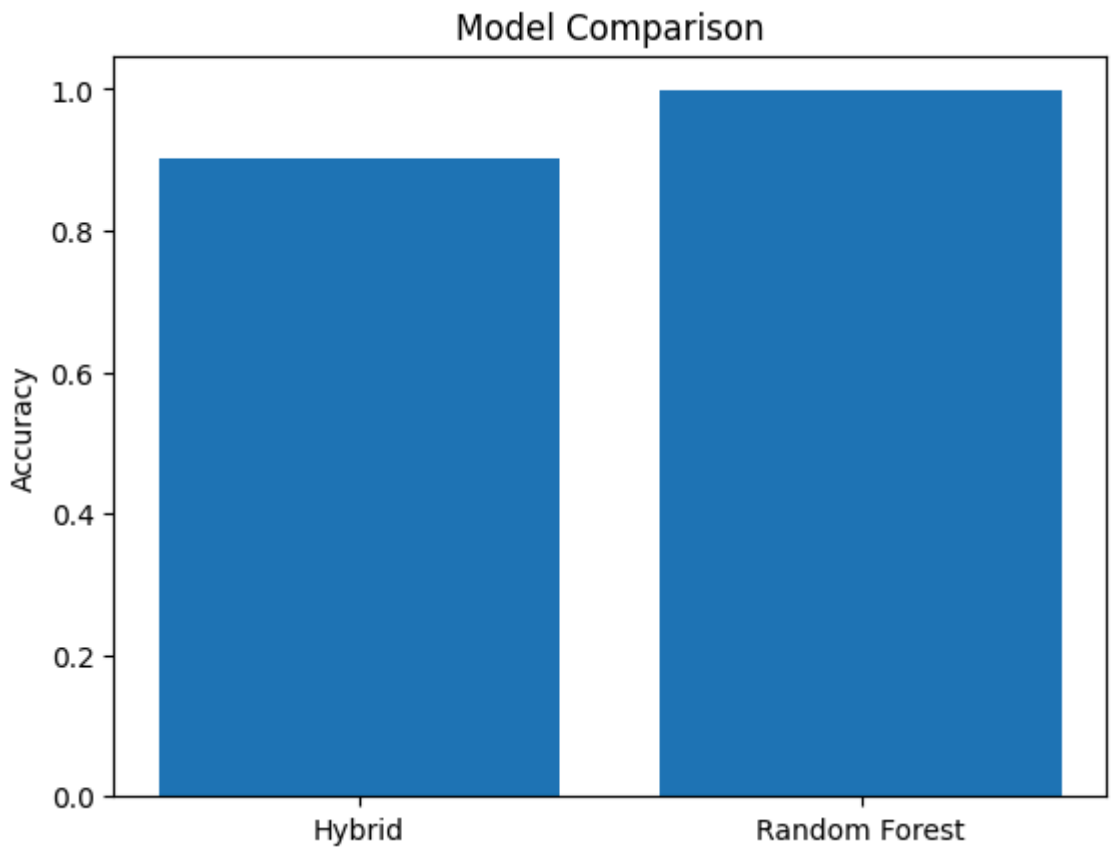
To evaluate the effectiveness of multi-modal learning, a Random Forest classifier is trained using only structured features:

f_RF(xs)

Performance comparison between:

Hybrid Model and Random Forest

provides insight into the contribution of textual modality and deep representation learning.



Final Interpretation

The proposed framework integrates structured symptom data, contextual information, and semantic textual embeddings into a unified representation optimized for multi-class disease classification. By combining nonlinear transformations, transfer learning, and feature-level fusion, the system is designed to simulate real-world clinical decision-making processes while remaining computationally efficient.